

An Intelligent Notification Urgency Classifier: TF-IDF/Naive Bayes Triage with Transformer-Based Summarization

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Abstract—Mobile users receive a continual stream of notifications, and the cognitive burden of manually triaging them by importance and urgency is a growing productivity problem. This paper describes an end-to-end machine learning and natural language processing (NLP) system that automatically classifies mobile notifications into the four quadrants of the Eisenhower Matrix (Urgent & Important, Important & Not Urgent, Urgent & Not Important, Not Urgent & Not Important) and additionally summarizes long notification text to help users quickly grasp critical information. The classification pipeline uses TF-IDF feature extraction with a Multinomial Naive Bayes classifier trained on a synthetic dataset of 440 professional notification samples (80/20 train/test split), and is compared against a rule-based keyword baseline. The trained classifier achieves 93% accuracy (macro F1 = 0.93, weighted F1 = 0.93), compared to 41% accuracy for the rule-based baseline, a 2.3× improvement in accuracy. A DistilBART transformer model (sshleifer/distilbart-cnn-12-6) is used for abstractive summarization of long notification messages. We describe the dataset construction, methodology, and evaluation results, and discuss the limitations of relying on a synthetic dataset along with directions for future work.

Index Terms—notification triage, text classification, TF-IDF, Naive Bayes, Eisenhower Matrix, abstractive summarization, BART, NLP

I. INTRODUCTION

Modern smartphone users receive hundreds of notifications a day from messaging apps, email clients, enterprise tools, and system alerts. This volume of interruption causes cognitive overload and makes it difficult for users to distinguish which notifications require immediate action from those that can be deferred or ignored. A well-known framework for prioritizing tasks by urgency and importance is the Eisenhower Matrix, which partitions items into four quadrants: Urgent & Important, Important & Not Urgent, Urgent & Not Important, and Not Urgent & Not Important.

This project applies the Eisenhower Matrix to the problem of notification triage by building a machine learning system that automatically classifies incoming notifications into one of these four quadrants. In addition to classification, the system integrates an AI summarization component so that long notification text (for example, a lengthy incident report or system alert) can be condensed into a short, actionable summary. The goal is to demonstrate that classical machine learning techniques, combined with modern transformer-based NLP, can meaningfully reduce the effort required to manage notification overload.

II. RELATED WORK

Text classification using TF-IDF (term frequency–inverse document frequency) feature representations combined with Naive Bayes classifiers is a long-standing and widely used approach in NLP, valued for its simplicity, interpretability, and strong performance on bag-of-words style classification tasks such as spam filtering and topic categorization. Multinomial Naive Bayes in particular is a standard baseline for document classification problems where features are word or n-gram counts.

For abstractive summarization, transformer-based sequence-to-sequence models have become the dominant approach. BART (Lewis et al.) is a denoising autoencoder pretrained for sequence generation tasks and has been fine-tuned for summarization on datasets such as CNN/DailyMail. DistilBART is a distilled, smaller version of BART designed to retain much of the summarization quality of the full model while being more efficient to run, making it practical for integration into lightweight application pipelines such as the one described in this paper.

This work does not propose new modeling techniques; rather, it combines an established classical text-classification pipeline with an established pretrained summarization model to address the applied problem of notification urgency triage.

III. DATASET

The system is trained on a synthetic dataset of professional notification messages constructed specifically for this project. The dataset consists of 440 samples, each labeled with one of the four Eisenhower Matrix quadrants (Urgent & Important, Important & Not Urgent, Urgent & Not Important, Not Urgent & Not Important). According to the project documentation, the dataset generation process includes deliberate injection of ambiguity to simulate the noise and overlap present in real-world notification text, rather than using cleanly separable synthetic examples.

The dataset is split into training and test sets using an 80/20 split, with the model trained on the 80% training partition and evaluated on the held-out 20% test partition.

IV. METHODOLOGY

A. Classification Pipeline

The core classification pipeline consists of two stages:

- 1) **Feature extraction:** Notification text is converted into TF-IDF vectors, which weight terms by their frequency in a given notification relative to their frequency across the full corpus, emphasizing terms that are distinctive to a given notification's content.
- 2) **Classification:** The TF-IDF feature vectors are passed to a Multinomial Naive Bayes classifier, which is trained to predict one of the four Eisenhower Matrix quadrant labels.

As a point of comparison, the system also implements a rule-based keyword classifier, which assigns urgency/importance labels based on the presence of predefined keywords in the notification text, without any learned model.

B. Summarization Module

For long notification messages, the system applies an abstractive summarization model, `sshleifer/distilbart-cnn-12-6`, a distilled variant of the BART transformer architecture. Given a long input notification (for example, a detailed server outage report), the model generates a condensed natural-language summary highlighting the most critical information, allowing a user to grasp the essential content without reading the full message.

C. Tech Stack

The system is implemented in Python using scikit-learn for the TF-IDF vectorization and Multinomial Naive Bayes classifier, and HuggingFace Transformers for the DistilBART summarization model. Pandas is used for data handling, and Matplotlib/Seaborn are used for visualization of results (confusion matrix and model comparison plots).

V. EVALUATION

A. Overall Classifier Performance

The Multinomial Naive Bayes classifier trained on TF-IDF features achieves an overall accuracy of 93% on the held-out test set, with a macro F1 score of 0.93 and a weighted F1 score of 0.93.

Figure 1 illustrates this overall accuracy in the context of the model comparison discussed in Section V-C.

B. Class-wise Performance

Table I reports precision, recall, and F1 score for each of the four Eisenhower Matrix classes, as documented in the project results. Figure 2 shows the corresponding confusion matrix on the held-out test set, illustrating which quadrants are most frequently confused by the classifier.

TABLE I
CLASS-WISE PRECISION, RECALL, AND F1 SCORE

Class	Precision	Recall	F1
Important & Not Urgent	1.00	0.88	0.93
Not Urgent & Not Important	0.81	1.00	0.89
Urgent & Important	0.93	1.00	0.97
Urgent & Not Important	1.00	0.84	0.91

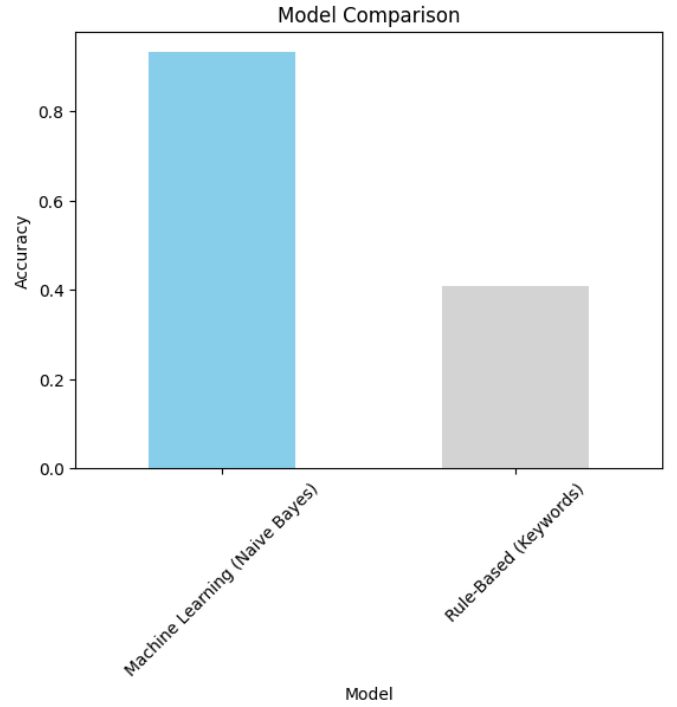


Fig. 1. Accuracy comparison between the TF-IDF/Multinomial Naive Bayes classifier and the rule-based keyword baseline.

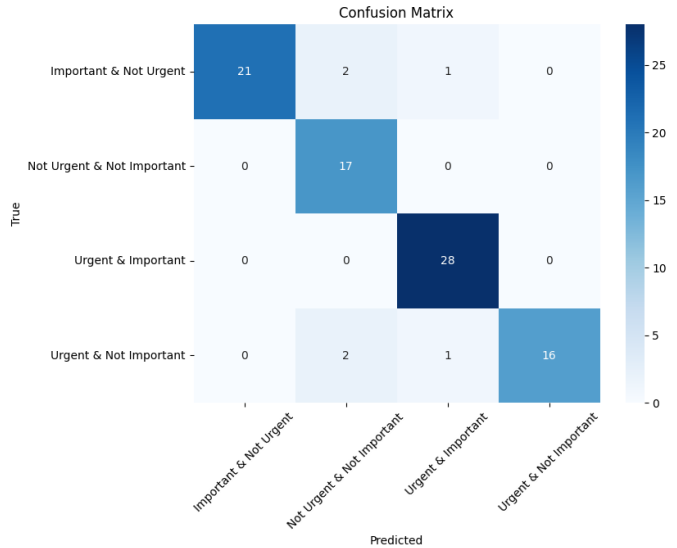


Fig. 2. Confusion matrix for the Multinomial Naive Bayes classifier on the held-out test set, broken down by Eisenhower Matrix quadrant.

C. Comparison Against Rule-Based Baseline

Table II compares the accuracy of the Multinomial Naive Bayes classifier against the rule-based keyword baseline. The machine learning classifier achieves 0.93 accuracy compared to 0.41 accuracy for the rule-based approach, corresponding to a $2.3\times$ improvement in accuracy over the heuristic baseline.

TABLE II
MACHINE LEARNING VS. RULE-BASED BASELINE ACCURACY

Model	Accuracy
Machine Learning (Naive Bayes)	0.93
Rule-Based Keywords	0.41

D. Summarization Example

The project documentation includes a qualitative example of the summarization module condensing a long server outage message into a short summary describing the root cause (a database connection pool exhausted by a memory leak) and the engineering team’s investigation, demonstrating the module’s ability to compress lengthy alert text into an actionable synopsis.

VI. LIMITATIONS

The primary limitation of this work is that the classifier is trained and evaluated entirely on a synthetic, professionally-styled notification dataset of 440 samples rather than on real, anonymized notification data. While the dataset generation process intentionally injects ambiguity to approximate real-world noise, synthetic text may not fully capture the diversity of phrasing, length, formatting, and domain-specific vocabulary found in actual mobile notifications from messaging apps, email clients, and system alerts. As a result, the reported 93% accuracy may not directly generalize to real-world deployment without further validation on authentic notification data. Additionally, the dataset size of 440 samples is relatively small for a four-class text classification task, and the TF-IDF representation, while effective on this dataset, does not capture contextual or semantic relationships between words in the way that contextual embedding models (e.g., BERT) would.

VII. CONCLUSION AND FUTURE WORK

This paper presented an end-to-end system for classifying mobile notifications into Eisenhower Matrix urgency/importance quadrants using a TF-IDF plus Multinomial Naive Bayes pipeline, combined with a DistilBART-based summarization module for condensing long notification text. On a synthetic dataset of 440 professional notification samples, the classifier achieved 93% accuracy, substantially outperforming a rule-based keyword baseline (41% accuracy), a $2.3\times$ improvement. These results suggest that even a classical, lightweight ML pipeline can substantially outperform simple heuristic rules for notification triage.

Documented directions for future work include replacing the synthetic dataset with real, anonymized notification data to validate generalization; deploying the model as a REST API for integration into mobile operating systems; replacing TF-IDF features with contextual embeddings such as BERT to better capture semantic meaning; and incorporating reinforcement learning to personalize notification prioritization to individual users over time. Potential real-world applications include smart notification filtering in mobile operating systems, productivity

and focus tools, email prioritization systems, enterprise alert triage platforms, and AI-driven personal assistants.